Workout Tracker

Product Requirements for an Android / Jetpack Compose Port

Document status: Implementation baseline

Purpose

This document captures the user-visible behavior of the existing Python/Kivy mobile application as requirements for a new native Android implementation. It is a behavioral specification, not a line-by-line translation. The target is an offline-first Jetpack Compose application with maintainable state, persistence, and integration boundaries.

Source baseline

The requirements were derived from **mobile_app/app.py**, **screens.py**, **widgets.py**, **kv.py**, **constants.py**, **utils.py**, and the supporting **common/db_helper.py** and **common/google_drive_helper.py**. Items marked **[Inferred]** make an Android-specific expectation explicit where the current code only implies it.

Product boundary

- The product is a personal workout log for recording dated workouts, exercises, sets, repetitions, and weights in kilograms.
- The primary user is an individual lifter using one local profile on one Android device.
- Core logging and browsing must work without a network connection; Google Drive is optional and user initiated.
- This port starts with a fresh Room database. Importing the legacy Kivy SQLite schema or its Drive backups is out of scope.

Terminology

Term	Meaning
Workout	A named training session on one calendar date.
Exercise entry	An occurrence of a catalog exercise within a workout.
Set	One repetitions-and-weight pair belonging to an exercise entry.
Exercise library	The reusable, case-insensitive catalog of exercise names, goals, and notes.
Goal	An optional target weight in kilograms for one catalog exercise.
Best set	The historically highest recorded weight for an exercise, with repetitions from that same set.
Draft	Unsaved new-workout editor state retained while moving between app destinations.

1. Navigation and Primary Views

NAV-01. The app shall expose persistent navigation to **Home**, **New Log**, **Goals**, and **Settings**. Workout detail is reached from Home and editing an existing workout is reached from detail.

NAV-02. Home shall return to an active workout's detail when applicable; otherwise it shall show the workout list. Saving an editor shall open the saved workout's detail.

NAV-03. Leaving and returning to a new-workout editor shall preserve its in-memory draft, including exercise order, set values, and expanded/collapsed exercise state. A draft is cleared after successful save or explicit clear.

Primary views

View	Required purpose and content
Workout list / Home	Search and browse workout cards showing name, formatted date, distinct exercise count, exercise-name preview, and an Open action.
Workout detail	Show workout name, formatted date, exercise-entry count, and every exercise with numbered rep x weight kg sets; provide edit and delete actions.
Workout editor / New Log	Create or edit a workout name/date and a dynamic ordered list of exercise entries and sets; expose workout and exercise notes.
Goals	List each catalog exercise with optional target, best set, percentage progress, and an Update Goal action.
Settings	Control theme, browse defaults, exercise library, Google Drive connection/backup/restore, and preference reset.

2. Workout Browsing

- BR-01. Search shall be case-insensitive and match substrings across workout name, stored ISO date, and exercise names.
- BR-02. Date filters shall cycle through All Time, Recent 30 days, Recent 90 days, and This Year. The comparison uses the device's current local date/time. Records with unparseable or missing dates remain visible under date filters, matching current behavior.
- BR-03. Sort shall cycle between Newest and Oldest. Date is the primary key and record identity resolves ties. Unknown dates sort as the earliest possible date.
- BR-04. Grouping shall cycle through None, Month, Year, and Workout Name. Unknown dates use an 'Unknown Date' section for date grouping. Changing grouping resets collapsed sections.
- BR-05. Group headers shall toggle expanded/collapsed state without losing the current search/filter/sort selection.
- BR-06. Browse preferences and search text shall persist across process restarts. Defaults are light theme, empty search, All Time, Newest, and no grouping.
- BR-07. The list shall initially load a date window of 14 days and append another window near the bottom. If remaining records have no valid dates, load at most 20 at a time. Continue loading until content can scroll.
- BR-08. Show a live count and the active filter, sort, and grouping labels. If nothing matches, show guidance to change the current browse controls.

3. Workout Creation and Editing

- ED-01. A new workout shall start with a blank name, today's local date, and at least one exercise entry with at least one set.
- ED-02. The date picker shall support valid calendar dates from 20 years before through 5 years after the current year, include a Today shortcut, display a readable preview, and save editor dates as DD-MM-YYYY.

- ED-03. Users shall add/remove exercise entries, select names from the exercise library, add/remove sets, and expand/collapse exercise cards. Adding a new exercise collapses existing cards to focus the new entry.
- ED-04. A collapsed exercise card shall summarize its set count and recorded rep/weight pairs.
- ED-05. Saving shall require a non-empty workout name, a valid DD-MM-YYYY or ISO YYYY-MM-DD date, at least one valid exercise entry, at least one set per exercise, and non-empty repetitions and weight for every set.
- ED-06. Repetitions shall accept whole-number input and weight shall accept decimal numeric input. **[Inferred]** Android validation shall reject negative values, non-finite values, and values that cannot be stored without loss.
- ED-07. Persisted dates shall use ISO YYYY-MM-DD; displayed dates shall use DD Mon YYYY; displayed sets shall use 'Set n: reps x weight kg'.
- ED-08. Saving a new exercise name shall add it to the reusable catalog. Editing a workout shall preserve its identity and atomically replace its exercise entries.
- ED-09. Clear and destructive actions shall require confirmation when entered content could be lost. Validation failures shall identify the first actionable field or set error and shall not partially save data.

4. Notes and Exercise Library

- LIB-01. Users shall manually add catalog exercises and see the current goal next to every catalog name.
- LIB-02. Exercise names shall be trimmed, non-empty, unique without regard to case, and sorted case-insensitively.
- LIB-03. Renaming a catalog exercise shall update matching historical workout entries and any open draft/editor selection.
- LIB-04. If a rename collides with another exercise, the app shall ask whether to combine them. Combining shall move historical entries to the target name, keep the larger of two non-null goals, prefer the target note when present, and remove the source catalog record.
- NOTE-01. Users shall create, update, clear, or close without changing multiline notes for a named workout template and for catalog exercises. A workout name or exercise selection is required before opening its note.
- NOTE-02. Workout notes are shared case-insensitively by workout name, not by individual workout ID. Exercise notes are shared by catalog exercise. The Android UI shall make this scope understandable. **[Inferred]**

5. Goals and Progress

- GOAL-01. The Goals view shall show every catalog exercise, including exercises without a target or workout history.
- GOAL-02. A goal is an optional decimal weight in kilograms; empty input clears it. **[Inferred]** Android shall require a finite value greater than zero when a goal is supplied.
- GOAL-03. Best set shall be the maximum historical weight across all aligned set pairs for the exercise, with repetitions from that selected set. With no history, show 0 reps x 0 kg.
- GOAL-04. Progress is best weight divided by goal times 100, displayed to one decimal place. If goal is null or zero, display N/A.
- GOAL-05. If the catalog is empty, show a helpful empty state describing how goals become available.

6. Preferences, Theme, and Responsive UI

- PREF-01. Users shall toggle light/dark themes and see the selected mode applied across screens, cards, inputs, buttons, text, dialogs, and scrims.
- PREF-02. Browse controls changed on Home or Settings shall share one persisted state. Reset Preferences shall restore theme and browse defaults, refresh visible content, and confirm completion.
- UI-01. The layout shall adapt at widths below approximately 430 dp while retaining all actions and readable labels.
- UI-02. Lists, dialogs, long notes, and long exercise/workout names shall remain usable without clipping; list changes shall return users to a predictable scroll position.
- UI-03. Empty states shall exist for no matching workouts, no workout exercises, no catalog exercises, and no goals.

7. Google Drive Backup and Restore

- DRV-01. Settings shall show availability, disconnected, authorizing, connected, uploading, restoring, and error states. Sign-in, backup, and restore actions shall disable while an operation is pending.
- DRV-02. Google authorization shall use an Android-native authorization flow with the minimum Drive scope needed for the app-managed backup. Sign-out shall clear locally held authorization state.
- DRV-03. Backup shall require explicit confirmation and upload the current complete Room database to a dedicated 'Workout Tracker Backups' Drive folder. Repeated backup shall update the same named file rather than create uncontrolled duplicates.
- DRV-04. Restore shall require a warning that local data will be overwritten, download the most recently modified matching Android backup into temporary storage, validate it, and replace local data atomically.
- DRV-05. A successful restore shall reopen persistence and refresh workout, detail, goal, and catalog state. Missing backups and service failures shall produce specific user-readable errors.
- DRV-06. Authentication or backup/restore failures that indicate invalid authorization shall disconnect the account and require a new sign-in. Temporary files shall be removed after every outcome.
- DRV-07. Network and file operations shall run outside the main thread and survive ordinary configuration changes without duplicated operations. **[Inferred]**
- DRV-08. Because this is a fresh Android data model, legacy Kivy backup files shall not be offered for restore.

8. Error and Confirmation Requirements

Situation	Required response
Workout missing	Show a not-found state; do not crash or expose stale detail.
Invalid editor field	Keep all input intact and identify the first field/set requiring correction.
Delete workout	Require confirmation, cascade-delete child exercise entries/sets, refresh lists, and return Home.

Situation	Required response
Clear workout or note	Require the appropriate explicit user action; clearing a note removes stored content.
Exercise name collision	Offer cancel or combine and describe the target exercise.
Concurrent Drive action	Ignore or disable duplicate actions until the active operation completes.
Restore invalid/corrupt	Keep or recover the prior local database; report failure and never leave persistence closed. [Inferred]

9. Android Non-Functional Requirements

Area	Requirement
Architecture	Use unidirectional data flow: immutable screen state from lifecycle-aware ViewModels, UI events into ViewModels, repositories as data boundaries, and dependency injection. Compose functions shall not access Room or Drive directly.
Offline-first	All workout, catalog, goal, note, draft, and preference behavior shall remain available without network access. Drive is an explicit secondary backup channel.
Lifecycle	Persist durable state in Room/DataStore; retain transient state with ViewModel and SavedStateHandle where appropriate. Do not lose input on rotation, backgrounding, or process recreation when restoration is feasible.
Data integrity	Use Room foreign keys, transactions, stable IDs, ordered child records, and normalized Workout, WorkoutExercise, and ExerciseSet rows. Never encode multiple sets in comma-separated fields.
Performance	Use observable Room queries and lazy Compose lists. Search/filter/group work must remain responsive for at least 10,000 workouts and must not block the main thread. [Inferred]
Accessibility	Provide meaningful semantics and content descriptions, 48 dp touch targets, scalable text, logical focus order, keyboard support, sufficient contrast, and non-color-only status indicators. [Inferred]
Security	Use Android account/authorization APIs; never package client secrets; store sensitive tokens through platform-protected facilities; use scoped app storage and TLS; exclude unintended data from device/cloud backup. [Inferred]
Reliability	Represent loading/success/empty/error states explicitly. Coroutines shall be cancellable, exceptions mapped to user-readable messages, and destructive replacements performed atomically.
Testability	Repositories and Drive gateways shall be interfaces with fakes. Business rules, Room queries/transactions, ViewModels, navigation, Compose semantics, and backup failure paths shall be automated.

10. Feature-to-Compose Traceability

Feature	Compose destination/component	Primary state owner
Browse/search/filter/sort/group/paging	WorkoutListScreen; WorkoutCard; BrowseControls; GroupHeader	WorkoutListViewModel / WorkoutListUiState
Workout details and deletion	WorkoutDetailScreen; ExerciseSummaryCard; ConfirmDeleteDialog	WorkoutDetailViewModel / WorkoutDetailUiState
Create/edit/draft/date/notes	WorkoutEditorScreen; ExerciseEditorCard; SetEditorRow; DatePicker; NoteDialog	WorkoutEditorViewModel / WorkoutEditorUiState
Goals and best-set progress	GoalsScreen; GoalCard; GoalDialog	GoalsViewModel / GoalsUiState
Theme and browse defaults	SettingsScreen; PreferenceControls; app theme	SettingsViewModel + PreferencesRepository
Exercise library/rename/combine	ExerciseLibraryDialog; ExerciseNameDialog; CombineDialog	SettingsViewModel + ExerciseRepository
Drive sign-in/backup/restore	DriveSettingsCard; ConfirmBackupDialog; ConfirmRestoreDialog	SettingsViewModel / BackupUiState
Navigation and active selection	WorkoutTrackerApp; NavHost; BottomNavigation	Navigation state + destination SavedStateHandle

11. Acceptance Summary

- A user can complete the full create-browse-detail-edit-delete workflow offline without losing valid data or transient editor input during ordinary navigation.
- Search, filters, sorting, grouping, pagination, notes, catalog changes, goals, best-set progress, theme, and preference reset reproduce the specified behavior.

- Room relationships and transactions preserve workout/exercise/set ordering and referential integrity.
- Drive operations are opt-in, visibly stateful, main-thread safe, confirmed when destructive, and recover safely on failure.
- Automated tests cover core validation/business rules, database behavior, ViewModel state transitions, critical Compose flows, and backup/restore errors.

Explicit assumptions

Units remain kilograms; the application remains single-user and local-first; there is no legacy-data import; user-visible current behavior is authoritative unless an [Inferred] Android requirement strengthens safety, accessibility, lifecycle handling, or validation.